

# Viyom Shukla

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🌐 Portfolio

## Education

Jaypee Institute of Information Technology, Noida

2022 – 2027

Integrated B.Tech + M.Tech in Computer Science Engineering – CGPA: 7.62/10

## Technical Skills

**Programming Languages:** C, C++, JavaScript, Python, HTML, CSS, SQL

**Frameworks & Libraries:** React.js, Node.js, Express.js, Tailwind CSS

**Tools & Technologies:** Git, GitHub, MongoDB, MySQL, API, VS Code, Notion, Slack

**Core Competencies:** Data Structures & Algorithms, Object-Oriented Design, Problem Solving, Complexity Analysis, Distributed Systems, Scalable Systems, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Data Mining

## Experience

Web Developer Intern

Aug 2025 – Oct 2025

*Distance Connect*

- Designed and developed scalable web application using React.js and Node.js in a multi-tier architecture.
- Built and integrated APIs enabling efficient data processing and real-time communication in distributed systems.
- Improved system performance and optimized frontend rendering, enhancing scalability and user experience by 40%.

## Publications

**Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification**

*IEEE Xplore Digital Library* 🔗

- Divided NSL-KDD dataset into 4 binary classification subsets (Normal vs DoS, Probe, R2L, U2R) for targeted attack detection in embedded network security systems.
- Generated synthetic data using Conditional GANs for minority attack classes (Probe, R2L, U2R) achieving 99.8% accuracy on imbalanced datasets.
- Integrated ensemble learning methods (Random Forest, SVM, Neural Networks) demonstrating optimal performance across different attack types for real-time network threat detection.

## Projects

Thumbblify: AI-Powered SaaS (Distributed System Integration)

*Source Code* 🔗

- Designed and implemented a scalable full-stack SaaS platform using MERN stack supporting 100+ users.
- Engineered a scalable MERN platform using APIs (Gemini, Replicate) for real-time, fault-tolerant image processing.
- Built a low-cost WhatsApp bot via Twilio API, enabling users to generate AI thumbnails via mobile.
- Developed a Chrome extension and MERN backend optimized for high-speed performance and incredible scale.

DeepMood: Depression Prediction Using GANs

*Notebook* 🔗

- Developed machine learning model using Generative Adversarial Networks (GANs) achieving 92% accuracy in depression prediction.
- Trained model on mental health datasets and generated synthetic data to address data scarcity in clinical datasets.

NextTrip: Tour Booking Management Platform

*Source Code* 🔗

- Engineered full-stack MERN application with role-based authentication for secure admin and user access.
- Designed admin dashboard for real-time tour management and booking analytics with MongoDB database integration.
- Designed database schema and optimized queries ensuring efficient storage and retrieval.

WalletWatcher: Expense Management Web App

*Live Link* 🔗

- Engineered a responsive MERN dashboard to visualize spending habits and manage personal budgets..
- Secured sensitive financial data using JWT-based authentication and industry-standard encryption.

## Achievements

- International Conference Speaker:** Presented research on Network Intrusion Detection and Network Security at 17th IC3-2025 at IIIT Noida Certificate 🔗
- Student Placement Committee Member:** Coordinating placement activities and supporting 200+ students in job search.
- Hackathon Experience:** Participated in premier national hackathons including Smart India Hackathon 2023, Adobe India Hackathon, Flipkart Grid 6.0, and Amazon ML Challenge, solving real-world problems.
- Concepts:** Experience solving 350+ DSA problems focusing on optimization, time complexity, and algorithm design.